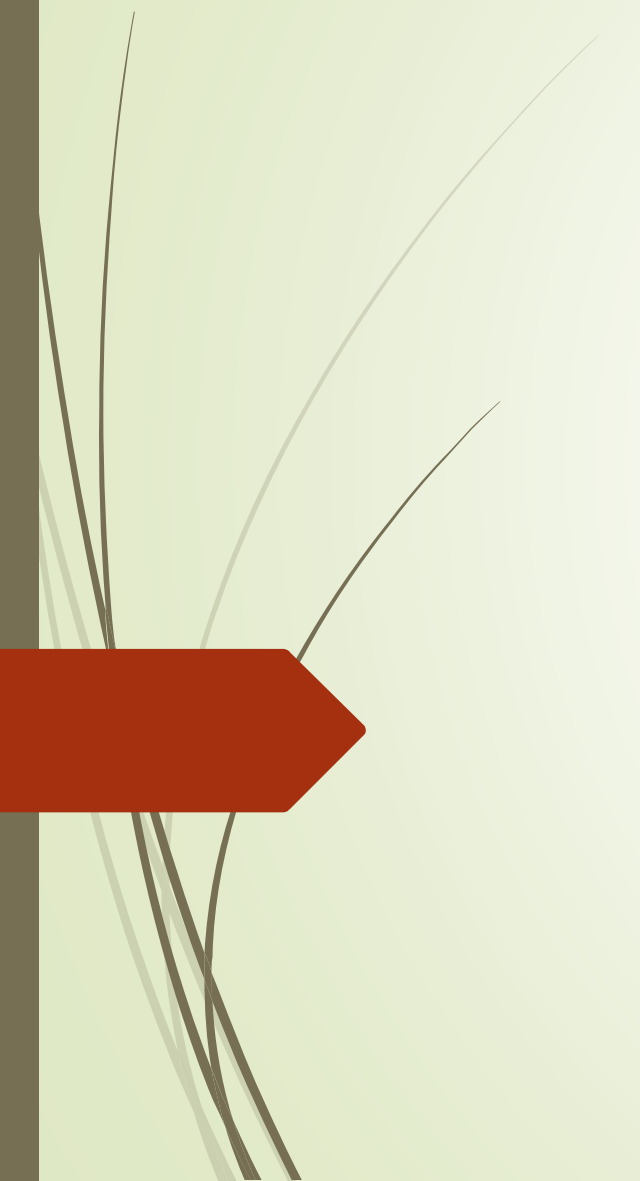




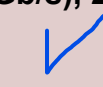
Literature review over
microring modulators.



■ contents

- **Comparison Table over Silicon Microring Modulator Papers.**

	Comparison Table — Silicon Microring Modulator Papers									
Device (paper)	Platform / process	λ (band)	Ring size	Waveguide cross-section	FSR (nm)	Loaded Q	Static ER (dB)	ER @ speed (dB @ rate)	data rate	Energy
Hu et al. — Cascaded double-ring, 2012.	0.18 μm CMOS on SOI.	C-b& ($\approx 1550\text{ nm}$)	two rings, 20 μm radius each.	bus/ring Rib WGs: 340 nm height, 80 nm slab; widths $\sim 450/500\text{ nm}$.	—	single-ring ≈ 5600 (0 V).	single $\approx 16\text{ dB}$; double $\approx 26\text{ dB}$.	3.9 dB @ 40 Gb/s.	40 Gb/s.	—
Xuan et al. — O-b& 15 μm ring, 2014.	OpSIS MPW SOI.	O-b& (1310 nm)	7.5 μm radius.	Strip-loaded: 420 nm width, 90 nm slab.	FSR $\approx 9.2\text{ nm}$.	Loaded Q ≈ 3500 .	—	6.2 dB @ 40 Gb/s NRZ-OOK.	40 Gb/s NRZ-OOK.	$\sim 115\text{ fJ/bit}$ estimated @ 4.8 Vpp.
Zhang et al. — 8 μm ring, optical (240 Gb/s), 2022.	220 nm SOI, CMOS foundry.	C-b& ($\approx 1550\text{ nm}$)	8 μm radius.	420 nm width, 70 nm slab thickness.	—	Q ≈ 4500 .	$>30\text{ dB}$.	—	120/220/240 Gb/s (with DSP).	—
Cai et al. — Lateral-zigzag PN, 9 μm ring, 2022.	SOI 220 nm.	C & O b&s tested	9 μm radius.	Ridge: 450 nm width, 90 nm slab.	—	Loaded Q ≈ 7000 .	—	5.9 dB@ 25 Gb/s NRZ; 3.9 dB@ 50 Gb/s; 1.7 dB@ 100 Gb/s PAM-4.	25/50 Gb/s NRZ, 100 Gb/s PAM-4.	$\sim 14\text{ fJ/bit}$ (100 Gb/s PAM-4)
Intel / Sakib et al. — 6 μm MRM (128 Gbaud), 2021.	—	O-b& ($\approx 1310\text{ nm}$)	6 μm radius.	Rib 300x400 nm; 100 nm slab thickness.	FSR $\approx 11.3\text{ nm}$.	Loaded Q ≈ 4200 .	—	$\sim 3.8\text{ dB}$ @128 Gbaud NRZ, 3.5 dB with no FFE ; PAM-4 192 Gb/s	128 Gbaud NRZ; 192 Gb/s PAM-4	$\sim 5.3\text{ fJ/bit}$ (device-level)

Device (paper)	PN geometry	Tunability (pm/V) or $V\pi\cdot L$	Insertion loss (dB)	Junction Cj	Series Rs	Electrical (RC) BW	Optical FWHM / Opt BW	Measured EO (S21) / 3-dB BW	Notes / key refs
Hu et al. — Cascaded double-ring, 2012.	Interleaved PN 600 nm length(300 nm for both P & N); 700 nm width, @ 2e17 cm ⁻³ doping.	$\Delta\lambda$ ~74 pm (double-ring); $V\pi\cdot L$ ~2.1–2.6 V·cm (case dep.).	4.4 dB	Cj ≈ 140 fF	Rs ≈ 32 Ω	21 GHz.	Single & cascaded FWHM ≈ 0.28, 0.41 nm → ~51 GHz opt BW.	—	Cascaded: adv. of a large optical b&width & a steeper sideb& with large ER.
Xuan et al. — O-b& 15 μm ring, 2014. 	Asymmetric lateral PN @ doping levels near 2e18 cm ⁻³ .	~14.2 pm/V; $V\pi\cdot L$ ≈ 2.2 V·cm.	7 dB under 4.8 Vp_p	Cj ≈ 20 fF @ -2.6 V.	Rs ≈ 70 Ω.	f_RC ≈ 43 GHz	FWHM 0.372 nm → ~65 GHz.	—	O-b& optimization with minimal degradation link advantage.
Zhang et al. — 8 μm ring, optical (240 Gb/s), 2022. 	Lateral PN, optimized doping (3e18 cm ⁻³).	$V\pi\cdot L$ = 0.825 V·cm (@4.5V).	12.5 dB, including 8 dB for grating couplers, 1.5 dB for optical fiber transmission, & 3 dB for bias loss.	Cj ≈ 38 fF.	Rj ≈ 11 Ω.	—	optical FWHM ~0.40 nm (~52 GHz).	52→110 GHz with optical peaking (detuning).	Optical peaking strategy; off-line DSP used at top rates.
Cai et al. — Lateral-zigzag PN, 9 μm ring, 2022. 	Lateral-zigzag PN (vertical center + 2 horizontal PNs).	~38 pm/V (70% doped), 54 pm/V fully doped; $V\pi\cdot L$ =0.76 V·cm @ 2.5V.	—	Cj ≈ 34 fF	Rs ≈ 27 Ω	f_RC ≈ 174 GHz.	—	~42 GHz.	high-efficiency with enlarged EO interaction area & reduced power consumption.
Intel / Sakib et al. — 6 μm MRM (128 Gbaud), 2021	—	$V\pi\cdot L$ ≈ 0.53 V·cm (3V bias).	—	Cj ≈ 33 fF	Rs ≈ 26 Ω	—	—	58–77 GHz (detuning dependent)	small radius, low energy/bit device-level, a threshold of BER<3.8e-3.

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References

- Xu, F.; Zhang, S.; "A High-Speed Silicon Ring Modulator with a Large Working Wavelength Range." *Electronics*, 2024, 13, 2890. DOI: 10.3390/electronics13152890.
- Moralis-Pegios, M.; Pitris, S; "52 km-Long Transmission Link Using a 50 Gb/s O-Band Silicon Microring Modulator Co-Packaged With a 1 V-CMOS Driver." *IEEE Photonics Journal*, Vol. 11, No. 4, Aug. 2019. DOI: 10.1109/JPHOT.2019.2921730.
- Zhang, Y.; Zhang, H "240 Gb/s Optical Transmission Based on an Ultrafast Silicon Microring Modulator." *Photonics Research*, 2022, 10, 1127.
- Cai, H.; Fu, S.; Yu, Y.; Zhang, X.
"Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator Working at 100 Gb/s." *IEEE Photonics Technology Letters*, 2022, 34, pp. 525–528. DOI: 10.1109/LPT.2022.3170554.
- Xu, Q.; Schmidt, B.; Shakya, J.; Lipson, M.
"Cascaded silicon micro-ring modulators for WDM optical interconnection." *Optics Express*, 2012, Vol. 14, pp. 9431–9435.



References

- Xuan, Z.; et al."Silicon microring modulator for 40 Gb/s NRZ-OOK metro networks in O-band." Optics Express, 2014, Vol. 22, pp. 28284–28291.
- Sakib, M.; Liao, P; "A High-Speed Micro-Ring Modulator for Next Generation Energy-Efficient Optical Networks beyond 100 Gbaud." CLEO / OSA conference contribution (postdeadline/Proc.), 2021..
- Müller, J., et al. "Optical peaking enhancement in high-speed ring modulators." Scientific reports 4.1 (2014): 6310.
- Ostrovskis, Armands, et al. "High-speed silicon photonics ring-resonator modulators for optical-amplification-free links." Optics Express 33.17 (2025): 36758-36769.